

Homework 5, Stat 3

1. Consider the Forbes2000 data in the R library HSAUR.
 - (a) For the variable category draw a bar chart.
 - (b) For the variable country draw a bar chart.
 - (c) Take the four categories that have the highest frequency and club every other category as others. Take the four countries that have the highest frequency and club every other country as others. Restrict your dataset to combinations of these (25 combinations)
 - (d) Draw a stacked bar plot with these 25 combinations and interpret the results.
2. Consider a logistic regression model with a single predictor and no intercept.
 - (a) What is the log likelihood of the slope parameter β .
 - (b) Write down the score equation, that is, find the derivative of the log likelihood and equate it to zero.
 - (c) Find the iterative equation of the Newton-Raphson method.
 - (d) Find the iterative equation of the Fisher scoring method.
3. Use the horseshoe crab data available here

`https://users.stat.ufl.edu/~aa/cda/data.html`

. Use Y as whether a female crab has a satellite.
 - (a) Use a single predictor X_1 as weight and fit a logistic regression model. Report the results.
 - (b) For the model in part (a), what is the probability that Y is one, when X is 5kg?
 - (c) Plot the fitted probabilities and confidence intervals for these probabilities as a function of X .
 - (d) Use two predictors, X_1 as weight and X_2 as width to fit a logistic regression. Report the results. Which predictors are significant at 5% level?